DSA Day 39: Merge Sort

Book: Data Structures Through C in Depth - Ch8 §8.12 · pp 434-444 | Code in Python

Overview

Divide the array, sort halves, merge. Stable, guaranteed O(n log n); ideal for linked lists.

Key concepts to master

- divide & merge
- stable
- extra O(n) space

Python implementation

```
def merge_sort(a):
    if len(a)<=1: return a
    m=len(a)//2
    L=merge_sort(a[:m]); R=merge_sort(a[m:])
    out=[]; i=j=0
    while i<len(L) and j<len(R):
        out.append(L[i] if L[i]<=R[j] else R[j])
        if L[i]<=R[j]: i+=1
        else: j+=1
    return out+L[i:]+R[j:]
```

Complexity

O(n log n) time, O(n) space.

Common pitfalls

- Losing stability with < instead of <=
- Extra allocations

Practice

- Sort {8,3,5,1,9,2}
- Why merge sort suits linked lists